

# Group Theory in a Nutshell for Physicists

## Chapter II.1 Exercises

1. Show that identity is in a class by itself.

the elements  $g, f \in G$  are in same class if

$$g = h^{-1} f h \quad \text{for any } h \in G$$

$$\Rightarrow h^{-1} I h = I h^{-1} h = I \quad \text{for all } h, I \in G$$

$\Rightarrow I$  is in a class by itself.

2. Show that in an abelian group, every element is in a class by itself.

Abelian group  $G$  has  $gh = hg$  for every  $g, h \in G$

$$\Rightarrow f^{-1} g f = g f^{-1} f = g I = g \quad \text{for all } f \in G \text{ \& } g \in G$$

So  $g$  is in a class by itself and  $g$  here represents every element of  $G$

$\Rightarrow$  Every element of Abelian group is in a class by itself.

3. ....

4. In chapter I.2, we worked out the equivalence classes of  $S_4$ . Calculate the characters of the 4-dimensional representation of  $S_4$  as function of its classes.

$S_4$  has 5 equivalence classes, with one of them being identity class

the character of class is given by:  $\chi^{(r)}(c) = \text{Tr } D^{(r)}(g)$  for  $g$  in class  $c$

So let's choose 1 representation of each class:

The class with only identity ( $c=1$ ) we get  $\left\{ \begin{array}{l} \text{4-d representation} \\ \text{means we have } D^{(4)}(g) \\ \text{let's suppose } (4) \end{array} \right.$

$$\chi(1) = \text{Tr } D(I) = \text{Tr } I_4 = 4$$

The class with only transpositions ( $c=2$ ) has the character

$$\chi(2) = \text{Tr } D((12)) = \text{Tr} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = 2$$

The class with 3-cycles ( $c=3$ ) has the character

$$\chi(3) = \text{Tr } D((123)) = \text{Tr} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = 1$$

The class with 4-cycles ( $c=4$ ) has the character

$$\chi(4) = \text{Tr } D((1234)) = \text{Tr} \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} = 0$$

The class with product of 2-cycles  $\uparrow$  ( $c=5$ ) has the character

$$\chi(5) = \text{Tr } (D((12))D((34))) = \text{Tr} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} = 0$$

